Leo Marcinkus

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LinkedIn | Engineering Portfolio | GitHub

EDUCATION

University of Colorado Boulder *Boulder, CO*

B.S. Electrical Engineering *Expected May 2027*

Major GPA: 3.66 | Cumulative GPA: 3.23 | Dean's List: Fall 2025, Spring 2026

M.S. Electrical Engineering *Expected May 2028*

EXPERIENCE

Siemens EDA, Software Test Intern (HyperLynx QA) *Boulder, CO | May 2026 - Present*

- Validate HyperLynx SI, PI, AMS, DRC, and Advanced Solver releases by running full simulation workflows on PCB designs, including **DDR4 and DDR5 timing margin**, eye diagrams, and crosstalk analysis with IBIS and IBIS-AMI models, and checking results for correctness.
- Run power integrity studies covering **DC drop, target impedance, decoupling, and VRM behavior**, and extract RLGC, current density, and EMI results with Fast 3D and Full-Wave solvers to verify tool output.
- Security-tested product logging against the EU Cyber Resilience Act requirement to record security-relevant activity, probing the software for ways to bypass or suppress logging and reporting the gaps found.
- Reproduce, isolate, and document software defects and workflow issues across releases in engineering reports, with usability recommendations for the development team.

Research Assistant, Photonic Neuromorphic Systems (Prof. Wagner) *CU Boulder | May 2026 - Present*

- Contributing to an in-pixel CMOS phase detector and liquid-crystal driver for a **25 x 25 µm pixel** on the X-FAB XS018 180 nm image-sensor process, where a dual-tap pinned photodiode and auto-zeroed comparator detect the phase of modulated light.
- Designing the in-pixel **1.8 V to 3.3 V level shifter** and latch that holds each pixel's comparator decision for a **full 1 ms frame** and drives the 3.3 V XOR liquid-crystal pad, balancing write margin, leakage, and pixel area.
- Set up the Siemens EDA custom IC flow (Custom IC, Calibre DRC/LVS) and MGLS licensing on a Rocky Linux VM for the group.
- Evaluate AI-assisted design studies against published work on comparator offset, kT/C and 1/f noise, and charge injection.

PROJECTS

Golden Arduino Board *Fall 2025*

- Designed and assembled a custom multi-layer ATmega328 board in Altium, interpreting multiple datasheets and benchmarking it against a stock Arduino Uno with oscilloscope measurements.
- Measured up to **1.7x lower ground bounce** and **1.9x better on-die supply isolation** than stock, from local decoupling placement and tighter return paths.
- Measured **2.2x better isolation** from external switching aggressors and **4.1x lower near-field emissions**, from tighter current loops and via stitching.

1T-1C DRAM Cell with Cross-Coupled CMOS Sense Amplifier *Spring 2026*

- **Designed, simulated, and laid out a 13-transistor two-cell DRAM array** in Siemens Tanner S-Edit, T-Spice, and L-Edit on a Generic 250 nm process at 2.5 V, including precharge, equalization, and sense amplifier enable timing.
- Measured a **57 mV pre-latch sense margin** against a 50 mV target, latched to a **1.5 V resolved differential in about 300 ps**, and verified destructive read and partial storage node refresh on both cells.

TECHNICAL SKILLS

Signal and Power Integrity: HyperLynx SI/PI/AMS and Advanced Solvers, Keysight ADS, SerDes Designer, Ansys Q2D and HFSS, IBIS/IBIS-AMI, DDR4/DDR5, S-parameters, eye diagrams, crosstalk, transmission lines, PDN and decoupling, transmit FFE

Test and Measurement: Oscilloscopes, TDR, VNA, logic analyzers, DMM, waveform generators, Digilent Analog Discovery, SMU I-V characterization

PCB and IC Design: Altium, KiCad, Cadence Virtuoso and Spectre, Tanner (S-Edit, T-Spice, L-Edit), Siemens Custom IC, Calibre DRC/LVS, LTspice

Embedded Systems: STM32 and ARM Cortex-M, FreeRTOS/CMSIS-RTOS2, Arduino, PWM, SPI/I2C/UART, RISC-V

Programming and Tools: C/C++, Python, MATLAB, SystemVerilog, Verilog-A, Tcl, Assembly, Git, Linux, AI-assisted workflows

INTERESTS

Avid collector of mechanical watches. Hands-on hardware hobbyist: custom mechanical keyboards and macropads (currently designing a custom RP2040 PCB), audio electronics, and 3D printing. Also play guitar and piano, skateboard, and follow the NBA.